CLASS 9 — SCIENCE · HARD LEVEL CHAPTER TEST SET 2

Chapter 4: **Journey Inside the Atom** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. An element has mass number 31 and 16 neutrons. The number of valence electrons is: 1

(a) 8 **(b)** 15 **(c)** 5 **(d)** 3

2. A nucleus with 82 protons and 125 neutrons has mass number: 1

(a) 43 **(b)** 125 **(c)** 207 **(d)** 82

3. Assertion (A): Sodium atom is more reactive than a sodium ion. Reason (R): The sodium ion has a complete octet. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

4. An element Z has 2 electrons in its K shell, 8 in L and 6 in M. The ion it is most likely to form is: 1

(a) Z^- **(b)** Z^{6+} **(c)** Z^{2-} **(d)** Z^{2+}

5. The ratio of the number of neutrons to protons is greatest in: 1

(a) tritium **(b)** helium-3 **(c)** protium **(d)** deuterium

6. Assertion (A): Neon is chemically inert. Reason (R): Its outermost shell is completely filled. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

7. The least number of neutrons is present in the atom of: 1

(a) oxygen-16 **(b)** hydrogen-1 **(c)** helium-4 **(d)** carbon-12

8. Which of the following has 8 neutrons? 1

(a) $^{12}_{6}C$ **(b)** $^{14}_{8}O$ **(c)** $^{14}_{7}N$ **(d)** $^{14}_{6}C$

9. **Assertion (A):** The mass number of an atom equals the number of protons plus the number of neutrons. 1
Reason (R): Electrons contribute most of the mass of an atom.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** A single chlorine atom has a mass of exactly 35.5 u. 1
Reason (R): Naturally occurring chlorine is a mixture of the isotopes ^{35}Cl and ^{37}Cl.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A particle has 9 protons, 10 neutrons and 10 electrons. Identify it with its symbol. 2

12. Find the number of electrons in SO_4^{2-} given S has 16 and O has 8 electrons. 2

13. An atom has 13 electrons. If it loses 3 electrons, find the charge, configuration and the noble gas it resembles. 2

14. State why neutrons are able to penetrate the nucleus more easily than alpha particles. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Calculate the average atomic mass of neon from isotopes 20 (90.5 per cent), 21 (0.3 per cent) and 22 (9.2 per cent). 3

16. Differentiate between an atom, an ion and an isotope using sodium as the example. 3

17. Why do metals form positive ions and non-metals negative ions? Use Mg and S. 3

OR

An atom has Z = 11, A = 23. Write the numbers of p, n, e in the atom, its ion with charge +1 and compare their sizes and reactivity. 3

18. The nucleus of an element X has 1 more proton than the nucleus of Y and the same mass number. Name the relationship and illustrate with an example. 3

OR

The relative abundance of isotopes of an element X-10 and X-11 is 1 : 4. Find the average mass and the element if Z = 5. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Boron occurs naturally as two isotopes: boron-10 (20 per cent) and boron-11 (80 per cent). Both have atomic number 5. A student has to find the average atomic mass of boron and explain why the two isotopes behave alike chemically. 5
(i) What are isotopes?
(ii) Find the number of neutrons in each isotope.
(iii) Find the average atomic mass of boron.
(iv) Why do isotopes have the same chemical properties, and name one use of a radioactive isotope?

20. Explain how elements of Z = 3 to 10 show the periodic variation of valency and the nature (metal, non-metal) with their configurations. 5

OR

Design a table for the number of protons, neutrons and electrons in $^{1}_{1}H$, $^{4}_{2}He$, $^{12}_{6}C$, $^{16}_{8}O^{2-}$, $^{23}_{11}Na^{+}$, $^{35}_{17}Cl^{-}$ and compute the totals for each. 5

END OF QUESTION PAPER — Total: 40 Marks